

4.)

Pokaži, da je $R = \begin{bmatrix} 1/2 & 1/2 & \frac{\sqrt{2}}{2} \\ 1/2 & 1/2 & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$ rotacijska. Poišči os in kot φ .

ortogonalna, $\det = 1$

$$RR^T = R^T R = I$$

$$RR^T = \frac{1}{4} \begin{bmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ -\sqrt{2} & \sqrt{2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & -\sqrt{2} \\ 1 & 1 & \sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} = I \checkmark$$

$$\det R = \frac{1}{2} \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{2}}{2} + \frac{\sqrt{2} \cdot \sqrt{2}}{2 \cdot 2 \cdot 2} + \frac{\sqrt{2} \cdot \sqrt{2}}{2 \cdot 2 \cdot 2} + \frac{\sqrt{2} \cdot \sqrt{2}}{2 \cdot 2 \cdot 2} = 1$$

$E_R(1)$:

$$(R - I)v = 0$$

$$\begin{bmatrix} -1/2 & 1/2 & \sqrt{2}/2 \\ 1/2 & -1/2 & -\sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -\sqrt{2} \\ -\sqrt{2} & \sqrt{2} & -1 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & -\sqrt{2} \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$z = 0$$

$$x = y$$

$$\vec{r} = x(1, 1, 0)$$

os rotacije

poljubna matrika za rot. v xy ! $R' = \begin{bmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\text{tr } R' = 2\cos \varphi + 1 = \text{tr } R = 1 \rightarrow \cos \varphi = 0, \quad \underline{\underline{\varphi = \frac{\pi}{2}}}$$

Rotacije v $\mathbb{R}^3$

- če $\det Q = 1$, je Q rot. mat. okoli osi skozi izhodišče

• $\sigma(Q)$: $\lambda_1 = 1$ $\lambda_{2,3} = e^{\pm i\varphi}$, $\varphi \in [0, \pi]$

• $\cos \varphi = \frac{\text{tr} Q - 1}{2}$

• $\cos = \frac{1}{2 \sin \varphi} \begin{bmatrix} Q_{32} - Q_{23} \\ Q_{13} - Q_{31} \\ Q_{21} - Q_{12} \end{bmatrix}$

Zrcaljenje v $\mathbb{R}^3$:

$\det Q = -1$, $\lambda_1 = -1$, $\lambda_{2,3} = 1$
 $\downarrow$
 normala ravnine zrc.

Zrcalni zasuki v $\mathbb{R}^3$:

$\det Q = -1$, ni zrcaljenje

• $\lambda_1 = -1$, $\lambda_{2,3} = e^{\pm i\varphi}$
 $\downarrow$ $\downarrow$
 os vrtenja kot vrtenja

(5.) Koliko je ort. transformacij $\mathbb{R}^3 \rightarrow \mathbb{R}^3$, ki preslikujejo

$\{x = \frac{y}{2} = z\}$ v $\{-\frac{x}{2} = y = z\}$, in

$\{x = -y = z\}$ v $\{x = y = z\}$?

še ena dimenzija v še ena dim.

B ortonormirana

B' ortonormirana

$B = [u_1 \ u_2 \ u_3]$

$B' = [v_1 \ v_2 \ v_3]$

$\rightarrow [Q]_{B'}^B = \begin{bmatrix} \pm 1 & 0 & 0 \\ 0 & \pm 1 & 0 \\ 0 & 0 & \pm 1 \end{bmatrix} \rightarrow 2^3 = 8$ možnosti

$Q(u_1) = \pm v_1$ $Q(u_3) = \pm v_3$

$Q(u_2) = \pm v_2$

Kvadratne forme

$$\underbrace{ax^2 + 2bxy + cy^2}_{\text{tip in orientacija}} + \underbrace{2dx + 2cy}_{\text{toži premik}} = \underbrace{f}_{\text{velikost / povečava}}$$

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad \begin{array}{l} \text{l. vekt.-orientacija} \\ \text{l. vr. - tip} \end{array}$$

1) Izračunaj glavne osi in skiciraj krivuljo

$$5x^2 - 6xy + 5y^2 = 8$$

$$A = \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix} \quad p_1(\lambda) = (5-\lambda)^2 - 9 = \lambda^2 - 10\lambda + 25 - 9 = \lambda^2 - 10\lambda + 16 = (\lambda-8)(\lambda-2)$$

$$E_A(8) = \text{Span}\{(1, -1)\}$$

$$\begin{bmatrix} -3 & -3 \\ -3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad x = -y \rightarrow (1, -1)$$

$$E_A(2) = \text{Span}\{(1, 1)\}$$

$$B = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = [id]_{\Sigma}^B$$

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$[v]_B = [id]_{\Sigma}^B [v]_{\Sigma}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = [id]_{\Sigma}^B \begin{bmatrix} x' \\ y' \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$x = \frac{1}{\sqrt{2}}(x' + y')$$

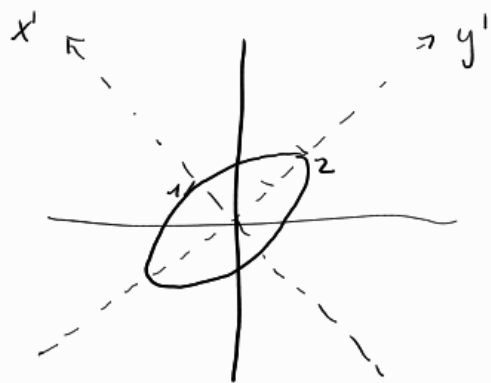
$$y = \frac{1}{\sqrt{2}}(-x' + y')$$

$$\frac{5}{2}(x' + y')^2 + \frac{5}{2}(-x' + y')^2 - 3(y'^2 - x'^2)$$

$$= \frac{5}{2}(x'^2 + 2x'y' + y'^2 + x'^2 - 2x'y' + y'^2) - 3y'^2 + 3x'^2 =$$

$$= 8x'^2 + 2y'^2 = 8$$

$$x'^2 + \frac{y'^2}{4} = 1$$



$$(2) \quad 2x^2 + 4xy - y^2 - \frac{12\sqrt{5}}{5}x - \frac{6\sqrt{5}}{5}y = 0$$

$$A = \begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix} \quad p_A(\lambda) = (\lambda - 3)(\lambda + 2)$$

$$E_A(-2) = \text{span}\{(1, -2)\}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix} \quad 2x = -y$$

$$x = x' + 2y'$$

$$E_A(3) = \text{span}\{(2, 1)\}$$

$$y = -2x' + y'$$

$$\begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix} \quad x = 2y$$

$$2(x' + 2y')^2 + 4(x' + 2y')(-2x' + y') - (-2x' + y')^2 - \frac{12\sqrt{5}}{5}(x' + 2y') - \frac{6\sqrt{5}}{5}(-2x' + y') =$$

$$= 2(x'^2 + 4x'y' + 4y'^2) + 4(-2x'^2 - 3x'y' + 2y'^2) - (4x'^2 - 4x'y' + y'^2) - \frac{12\sqrt{5}}{5}(x' + 2y') - \frac{6\sqrt{5}}{5}(-2x' + y') =$$

$$= -10x'^2 + 15y'^2 + x' \left(\frac{12\sqrt{5}}{5} - \frac{12\sqrt{5}}{5} \right) + y' \left(-\frac{24\sqrt{5}}{5} - \frac{6\sqrt{5}}{5} \right) =$$

$$= -10x'^2 + 15y'^2 - 6\sqrt{5}y' = 0$$

$$-10x'^2 + 15y'^2 - \frac{2\sqrt{5}}{25}y' + \frac{1}{5} = \frac{1}{5} \cdot 15 = 3$$

$$-10x'^2 + 15\left(y' + \frac{1}{15}\right)^2 = 3$$

